

Zhentao He

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github.com/JimmyTauH

EDUCATION

Fudan University

Ph.D.

Shanghai, China

Sept. 2025 - June 2030

Wuhan University

B.E. in Computer Science & Technology

Wuhan, Hubei Province, China

Sept. 2021 - June 2025

- Hongyi Honor College Student
- National Scholarship (2023-2024), Outstanding Graduate Award
- GPA: 3.91/4.00, IELTS Band 7

PUBLICATIONS

- Interpretable Data Integration for Single Cell and Spatial Multi-Omics. **Cell Systems** (2026).
Chenghui Yang, **Zhentao He**, Qing Nie and Lihua Zhang [Link](#)
- MoE-TCR: Mixture-of-experts framework for pan-specific TCR-epitope binding prediction. **Pattern Recognition** (2026).
Wei Qu, Jinxing Li, **Zhentao He**, Jiayi Wang and Shanfeng Zhu [Link](#)
- SpaDC enables sequence-based integrative analysis and regulatory inference of spatial chromatin accessibility data. **Communications Biology** (2026).
Chuanlong Ma, Chenghui Yang, Caiwei Zhen, **Zhentao He**, Yong Luo and Lihua Zhang [Link](#)

RESEARCH INTERESTS

Machine Learning and Data Mining for Biological Data

Developing robust, interpretable, and transferable representations for multiomics analysis, perturbation modeling, and protein function.

Representation Learning; Generalization & Adaptation ;

Multimodal Learning for Multiomics ; Perturbation Prediction & Protein Function Modeling

EXPERIENCE

PhD Student Researcher

Lab of Data Mining and Intelligent Information Processing, ISTBI, Fudan University

Sept. 2025 - June 2030

Advisor: Prof. Shanfeng Zhu

- Integration and analysis of spatial multiomics data.

Undergraduate Researcher

[BIOD Lab](#), School of Computer Science, Wuhan University

July 2022 - June 2025

Advisor: Prof. Lihua Zhang

- Develop methods single-cell multiomics data integration and analysis.
- Predict the impact of drug and genetic perturbations on cell states.

iGEM Team Member(Head of Model Team)

[WHU-China](#), CRISPRreporter

Nov. 2022 - Nov. 2023

Advisor: Dr. Lei Bai, Prof. Zhixiong Xie & Prof. Lihua Zhang

- **Gold Medal**; Nomination for Best Foundational Advance Project, etc.
- A novel multi-level DNA memory device with CRISPR technology and **its software suite**. [Description](#)

COMPETENCIES & SKILLS

Programming Languages: Python, C++, R

Libraries and Tools: PyTorch, Git, Docker; Scanpy, Seurat, etc.

ML Architectures: VAE, GNN, GAN, OT(Neural Optimal Transport), Transformer, etc.